

## Graphical Abstract

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## Highlights

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- Research highlight 1

- Research highlight 2

# Spectral NFST for Scalable Multiclass Novelty Detection in IoT Systems

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## Abstract

**Keywords:** Anomaly Detection, Controlled-Point Projection, Density Stratification, Kernel Methods, Discriminant Analysis.

## 1. Preliminaries: Null Foley-Sammon Transform

Null Foley-Sammon Transform (NFST) has emerged as a robust discriminative framework, specifically engineered to circumvent the Small Sample Size (SSS) problem in high-dimensional settings. NFST identifies a subspace where the within-class scatter completely vanishes while simultaneously preserving the positive variance of the total scatter.

*Scatter Matrices.* Consider a training set  $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$  belonging to  $c$  classes  $\{C^{(1)}, \dots, C^{(c)}\}$ . Let  $n_j$  be the number of samples in class  $C^{(j)}$ . The class mean and global mean are defined as:

$$\boldsymbol{\mu}_j = \frac{1}{n_j} \sum_{\mathbf{x}_i \in C^{(j)}} \mathbf{x}_i, \quad \boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i. \quad (1)$$

The within-class scatter  $\mathbf{S}_w$  and between-class scatter  $\mathbf{S}_b$  are:

$$\mathbf{S}_w = \frac{1}{n} \sum_{j=1}^c \sum_{\mathbf{x}_i \in C^{(j)}} (\mathbf{x}_i - \boldsymbol{\mu}_j)(\mathbf{x}_i - \boldsymbol{\mu}_j)^T, \quad (2)$$

$$\mathbf{S}_b = \frac{1}{n} \sum_{j=1}^c n_j (\boldsymbol{\mu}_j - \boldsymbol{\mu})(\boldsymbol{\mu}_j - \boldsymbol{\mu})^T. \quad (3)$$

The total scatter matrix is  $\mathbf{S}_t = \mathbf{S}_w + \mathbf{S}_b$ . An NFST projection vector  $\boldsymbol{\theta}$  must satisfy:

$$\boldsymbol{\theta}^T \mathbf{S}_w \boldsymbol{\theta} = 0, \quad \boldsymbol{\theta}^T \mathbf{S}_t \boldsymbol{\theta} > 0, \quad (4)$$

i.e. it lies in the null space of  $\mathbf{S}_w$  while staying outside the null space of  $\mathbf{S}_t$ .

*Algorithmic Derivation.* We first address the positivity of total scatter variance ( $\boldsymbol{\theta}^T \mathbf{S}_t \boldsymbol{\theta} > 0$ ) by ensuring the projection vector  $\boldsymbol{\theta}$  lies within the range of the total scatter matrix. This is achieved by representing  $\boldsymbol{\theta}$  as a linear combination of the basis vectors of the centered data:

$$\boldsymbol{\theta} = \mathbf{Q}\boldsymbol{\alpha}. \quad (5)$$

where  $\mathbf{Q} = [\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_{n-1}]$  is an orthonormal basis for  $\text{span}\{\mathbf{x}_i - \boldsymbol{\mu} \mid i = 1, \dots, n\}$ . By choosing  $\boldsymbol{\theta}$  from this span, we guarantee that  $\boldsymbol{\theta}^T \mathbf{S}_t \boldsymbol{\theta} > 0$  for any non-zero vector  $\boldsymbol{\alpha}$ , thus preventing the loss of discriminative information.

Next, for vanishing of between scatter variance ( $\boldsymbol{\theta}^T \mathbf{S}_w \boldsymbol{\theta} = 0$ ), we substitute the representation  $\boldsymbol{\theta} = \mathbf{Q}\boldsymbol{\alpha}$  into the within-class scatter condition:

$$(\mathbf{Q}\boldsymbol{\alpha})^T \mathbf{S}_w (\mathbf{Q}\boldsymbol{\alpha}) = 0 \implies \boldsymbol{\alpha}^T (\mathbf{Q}^T \mathbf{S}_w \mathbf{Q}) \boldsymbol{\alpha} = 0. \quad (6)$$

This condition is equivalent to finding the null space of the projected within-class scatter matrix. We solve the following system for  $\boldsymbol{\alpha}$ :

$$(\mathbf{Q}^T \mathbf{S}_w \mathbf{Q}) \boldsymbol{\alpha} = 0. \quad (7)$$

Solving this system yields  $c - 1$  optimal vectors  $\boldsymbol{\alpha}_i$ . The final projection vectors are recovered via  $\boldsymbol{\theta}_i = \mathbf{Q}\boldsymbol{\alpha}_i$ , forming the transformation matrix

$$\boldsymbol{\Theta} = [\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots, \boldsymbol{\theta}_{c-1}]. \quad (8)$$

## 2. Spectral NFST

The proposed Spectral NFST framework is designed to handle complex, multi-modal distributions by performing class-specific manifold decomposition. The

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process ensures that the internal structure of each known class is preserved before unified null-space learning. The methodology proceeds in the following phases:

*Phase 1: Kernel Mapping.* The objective of this phase is twofold: (i) to project the input data into a high-dimensional feature space where non-linear manifold structures become linearly separable, and (ii) to ensure the linear independence of the sample representations, which is a fundamental prerequisite for NFST.

Given a set of input observations  $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset \mathbb{R}^d$ , we employ a non-linear, strictly positive definite (SPD) kernel function  $k: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$  (e.g., the Gaussian Radial Basis Function). This kernel induces an implicit mapping  $\phi: \mathbb{R}^d \rightarrow \mathcal{H}$  into a Reproducing Kernel Hilbert Space (RKHS). To render this mapping computationally tractable and explicitly defined, we construct the kernel matrix  $\mathbf{K} \in \mathbb{R}^{n \times n}$ , where  $\mathbf{K}_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ . Then we treat each column of  $\mathbf{K}$  as a high-dimensional feature vector, defined as a  $\mathbf{z}_j \in \mathbb{R}^n$ :

$$\mathbf{z}_j = [k(\mathbf{x}_1, \mathbf{x}_j) \quad k(\mathbf{x}_2, \mathbf{x}_j) \quad \dots \quad k(\mathbf{x}_n, \mathbf{x}_j)]^T \quad (9)$$

*Phase 2: Targeted Manifold Decomposition.* To preserve the multi-modal structure of each category, we perform a class-specific manifold decomposition. For each class  $C^{(k)}$ , we construct a local affinity graph  $G^{(k)}$  where edge weights are defined by the kernel similarity values  $\mathbf{K}_{ij}$ . We then extract the Maximum Spanning Tree (MaxST) as the topological skeleton of the class, prioritizing the strongest mutual similarity links.

To achieve exactly  $Q_k$  subgroups, we remove the  $Q_k - 1$  edges with the lowest similarity values in the MaxST. This process systematically decouples the manifold at its weakest correlation points, yielding disjoint components  $\{C_1^{(k)}, \dots, C_{Q_k}^{(k)}\}$  that maximize internal cohesion.

*Phase 3: Global Component Relabeling.* To bridge the class-wise decomposition with the unified scatter analysis, we re-index each discovered subgroup into a global set  $\{C_k\}_{k=1}^Q$ , where  $Q = \sum_{k=1}^c Q_k$ . Specifically, a sample belonging to the  $l$ -th sub-manifold of class  $C^{(j)}$  is assigned a unique global component index  $k = (\sum_{m=1}^{j-1} Q_m) + l$ .

*4: Indicator Matrix Formalization.* These  $Q$  disjoint components are consolidated into the Global Indicator Matrix  $\mathbf{H} \in \mathbb{R}^{n \times Q}$ , which provides an orthogonal mapping for the similarity scatter analysis:

$$H_{ik} = \begin{cases} 1/\sqrt{|C_k|} & \text{if sample } i \in C_k \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

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#### Algorithm 1 Spectral NFST: Multiclass Training Phase

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**Require:**  $\{X_j\}_{j=1}^c$  (samples per class),  $\{Q_j\}_{j=1}^c$  (target components),  $k(\cdot, \cdot)$ .

**Ensure:**  $\Theta, \{m_k\}_{k=1}^Q$ .

- 1:  $\mathbf{K} \in \mathbb{R}^{n \times n} \leftarrow K_{uv} = k(x_u, x_v)$ .
  - 2: **for**  $j = 1$  to  $c$  **do**
  - 3:    $G^{(j)} \leftarrow$  Affinity graph of  $C^{(j)}$ .
  - 4:    $\{C_l^{(j)}\}_{l=1}^{Q_j} \leftarrow \text{MaxST}(G^{(j)}) \setminus \{Q_j - 1 \text{ weakest edges}\}$ .
  - 5: **end for**
  - 6:  $Q = \sum_{j=1}^c Q_j$ .
  - 7: Relabel  $\{C_l^{(j)}\} \rightarrow \{C_k\}_{k=1}^Q$  via index  $k = (\sum_{m=1}^{j-1} Q_m) + l$ .
  - 8:  $\mathbf{H} \in \mathbb{R}^{n \times Q} \leftarrow H_{ik} = 1/\sqrt{|C_k|}$  if  $i \in C_k$ , else 0.
  - 9:  $\mathbf{S}_w = \mathbf{K}(\mathbf{I} - \mathbf{H}\mathbf{H}^T)\mathbf{K}$ .
  - 10:  $\mathbf{S}_b = \mathbf{K}(\mathbf{H}\mathbf{H}^T - \frac{1}{n}\mathbf{1}\mathbf{1}^T)\mathbf{K}$ .
  - 11:  $\mathbf{V} \leftarrow$  Orthonormal basis for range of centered  $\mathbf{K}$ .
  - 12: Solve  $(\mathbf{V}^T \mathbf{S}_w \mathbf{V})\alpha = 0$  for  $\{\alpha_i\}_{i=1}^{Q-1}$ .
  - 13:  $\Theta = \mathbf{V}[\alpha_1, \dots, \alpha_{Q-1}]$ .
  - 14:  $m_k = \Theta^T \mu_k^{\text{sim}}$  where  $\mu_k^{\text{sim}} = \frac{1}{|C_k|} \sum_{i \in C_k} z_i, \forall k \in \{1, \dots, Q\}$ .
  - 15: **return**  $\Theta, \{m_k\}_{k=1}^Q$
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*Phase 5: Null Projection.* By leveraging the indicator matrix  $\mathbf{H}$ , we generalize the scatter analysis to the similarity space. Then the within and between scatter matrices become

$$\mathbf{S}_w = \mathbf{K}(\mathbf{I} - \mathbf{H}\mathbf{H}^T)\mathbf{K} \quad (11)$$

$$\mathbf{S}_b = \mathbf{K}(\mathbf{H}\mathbf{H}^T - \frac{1}{n}\mathbf{1}\mathbf{1}^T)\mathbf{K} \quad (12)$$

Following NFST, we obtain  $Q - 1$  projection vectors forming  $\Theta \in \mathbb{R}^{n \times (Q-1)}$ .

### 3. Implement Algorithm

Based on the theoretical derivation presented above, the training procedure for obtaining the transformation matrix  $\Theta$  and the projected sub-group centroids  $\{\mathbf{m}_j\}_{j=1}^Q$  is formally detailed in Algorithm 1.

#### 3.1. Multiclass Classification

During the inference phase, for any test sample  $\mathbf{x}_{\text{test}}$ , we first construct its similarity profile  $\mathbf{k}_{\text{test}} = [k(\mathbf{x}_1, \mathbf{x}_{\text{test}}), \dots, k(\mathbf{x}_n, \mathbf{x}_{\text{test}})]^T$  and embed it into the learned discriminative null space as  $\mathbf{y}_{\text{test}} = \Theta^T \mathbf{k}_{\text{test}}$ . This scoring procedure, which identifies anomalies based on their distance to the manifold backbone, is detailed in Algorithm 3.

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**Algorithm 2** Spectral NFST: Inference Phase

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**Require:**  $x_{test}, \mathcal{X}, \Theta, \{m_k\}_{k=1}^Q$ .

**Ensure:** Predicted class label  $\hat{y} \in \{1, \dots, c\}$ .

- 1:  $k_{test} \in \mathbb{R}^n \leftarrow [k_{test}]_i = k(x_i, x_{test}), \forall i \in \{1, \dots, n\}$ .
  - 2:  $y_{test} = \Theta^\top k_{test}$ .
  - 3:  $k^* = \arg \min_{k \in \{1, \dots, Q\}} \|y_{test} - m_k\|_2$ .
  - 4: Let  $L(k)$  be the original class label associated with sub-group  $C_k$ .
  - 5:  $\hat{y} = L(k^*)$ .
  - 6: **return**  $\hat{y}$
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### 3.2. Multiclass Novelty Detection

The inference phase of Spectral NFST is designed to address the dual challenge of multiclass classification and novelty detection. Unlike traditional discriminative models that are limited to closed-set scenarios, our framework utilizes the zero-variance properties of the null space to establish a rigorous decision boundary for unseen observations.

For an incoming test sample  $\mathbf{x}_{test}$ , the model first identifies its proximity to the manifold backbone by searching for the nearest projected sub-group centroid  $\mathbf{m}_{k^*}$  in the discriminative subspace. The degree of novelty is quantified by the Euclidean distance  $\mathcal{A}$ , which serves as a metric for manifold deviation. By introducing a decision threshold  $\tau$ , the model effectively partitions the space: if the distance  $\mathcal{A}$  exceeds  $\tau$ , the sample is flagged as a *Novelty*, indicating a lack of intrinsic similarity to any known attack or normal profile. Otherwise, the sample is assigned the original class label  $L(k^*)$  associated with the closest sub-manifold. This unified decision logic is formally detailed in Algorithm 3.

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**Algorithm 3** Spectral NFST: Multiclass Novelty Detection Phase

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**Require:**  $\mathbf{x}_{test}, \mathcal{X}, \Theta, \{\mathbf{m}_k, L(k)\}_{k=1}^Q$ , threshold  $\tau$ .

**Ensure:** Predicted label  $\hat{y}$  and anomaly score  $\mathcal{A}$ .

- 1: {Empirical Similarity Mapping}
  - 2:  $\mathbf{k}_{test} \in \mathbb{R}^n \leftarrow [\mathbf{k}_{test}]_i = k(\mathbf{x}_i, \mathbf{x}_{test}), \forall i \in \{1, \dots, n\}$ .
  - 3: {Subspace Embedding and Distance Computation}
  - 4:  $\mathbf{y}_{test} = \Theta^\top \mathbf{k}_{test}$ .
  - 5:  $k^* = \arg \min_{k \in \{1, \dots, Q\}} \|\mathbf{y}_{test} - \mathbf{m}_k\|_2$ .
  - 6:  $\mathcal{A} = \|\mathbf{y}_{test} - \mathbf{m}_{k^*}\|_2$ .
  - 7: {Novelty-aware Classification Logic}
  - 8: **if**  $\mathcal{A} > \tau$  **then**
  - 9:    $\hat{y} = \text{Novelty (or } -1)$ .
  - 10: **else**
  - 11:    $\hat{y} = L(k^*)$  (Original class label).
  - 12: **end if**
  - 13: **return**  $\hat{y}, \mathcal{A}$
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